Bank Telemarketing Campaign Prediction: Predicting Term Deposit Subscriptions Using a Stacked Ensemble Team 13

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Introduction

Telephone-based telemarketing is a primary customer acquisition channel for retail banks, yet subscription rates rarely exceed 10–15%, meaning most calls are wasteful (Xie et al. 2023). During Portugal's post-2008 financial crisis, banks were compelled to expand their deposit bases to meet capital adequacy requirements (Moro, Cortez, and Rita 2014), making efficient prospect targeting a strategic necessity. Machine learning (ML) offers a principled solution: score customers by predicted subscription likelihood and allocate a finite call budget to the highest-ranked prospects.

We address this problem on a dataset of $\sim$41,000 records using a Stacked Generalization ensemble evaluated under nested 5$\times$5 cross-validation. Our key contributions are: (i) a **leakage-free** pipeline that excludes call duration to ensure prospective deployability; (ii) a **dual preprocessing track** tailored to each model's assumptions; (iii) **nested CV** for unbiased hyperparameter selection; and (iv) results translated into **profit-maximising business metrics** grounded in Portuguese banking economics.

Related Work. Moro et al. (Moro, Cortez, and Rita 2014) established the UCI Bank Marketing benchmark, reporting AUC ≈ 0.80 with neural networks, recovering 79% of subscribers from the top-scoring half. Xie et al. (Xie et al. 2023) showed a hybrid Random Subspace–Multi-Boosting ensemble outperformed standalone learners, identifying contact month and job type as key drivers via Partial Dependence Plots. Peter et al. (Peter et al. 2025) demonstrated stacking ensembles reaching AUC 0.9491. Three gaps persist across this literature: (1) call duration is widely included despite introducing data leakage; (2) standard (non-nested) CV creates optimism bias during hyperparameter tuning; and (3) model results are rarely translated into actionable business value.

Dataset and Preprocessing

The dataset contains 41,188 records across 20 features: demographics (*age, job, marital, education, default, housing, loan*), campaign contact (*contact, month, day_of_week, campaign, previous, pdays*), one leaky feature (*duration*), and five macroeconomic indicators (*emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed*). The binary target records term deposit subscription (positive rate: 11.27%; imbalance ratio 7.88:1), necessitating class-imbalance-aware metrics and modelling.

Data Quality. Missing values are encoded as `unknown` strings and retained as a distinct category, as missingness may carry signal (Bhuva 2025). VIF analysis revealed severe multicollinearity: *euribor3m* (VIF 64.5), *emp.var.rate* (33.1), *nr.employed* (31.8). The feature *pdays* encodes never-contacted clients as 999 (96.32% of records). The dataset contains 12 duplicate rows (0.03%), removed during preprocessing.

Preprocessing. After removing 12 duplicate rows (0.03%), *duration* was excluded to prevent data leakage, as it is only known post-contact and highly correlates with outcomes. Missing values encoded as -1 in *housing*, *loan*, and *default* were converted to "unknown" categories, preserving genuine uncertainty as a predictor. A binary indicator *contacted_before* was engineered from *pdays*, capturing contact history more interpretably; *pdays* itself (VIF $\approx$ 15) and three highly collinear macroeconomic variables (*emp.var.rate*, *nr.employed*, *cons.price.idx*; VIF $>$ 10) were then dropped for linear models, retaining *euribor3m* and *cons.conf.idx* as economic signals.

Methods

To predict whether a client will subscribe to a term deposit or not based on the data provided, we first considered a Logistic Regression model. As a generalized linear model, Logistic Regression is best suited for binary classification, providing a probabilistic output $\mathrm{P}(y = 1|x)$ rather than a hard label (Kakrani 2024). This probability output allows us to maximise the recall of the positive class by adjusting the classification threshold, even at the risk of higher false positives; a trade-off that aligns with the bank's goal of minimizing missed opportunities. Furthermore, the coefficients of the Logistic Regression model allow for clear feature importance analysis, identifying the demographic and socioeconomic factors most predictive of a successful campaign for banks to focus on .

We formulate binary classification: given features $\mathbf{x}$, predict $\hat{y} \in \{0, 1\}$ indicating subscription. Two preprocessing tracks handle model-specific requirements. For **tree-based learners** (RF, XGB), categorical features are encoded via mean target encoding, replacing categories with their mean target value computed on the training fold. For **linear and distance-based learners** (LR, KNN, NB), binary encoding is applied followed by `StandardScaler` normalisation. All transformation parameters are fitted solely on training data within each fold, ensuring no information leakage.

Stacked Generalization. Four diverse base learners (Level-0) are combined by a Logistic Regression meta-classifier (Level-1); see Figure 1. **Random Forest** reduces variance through bagging over decision trees, with `class_weight='balanced'` to address class imbalance. **XGBoost** applies gradient boosting with `scale_pos_weight`, capturing non-linear interactions. **KNN** provides a non-parametric, instance-based perspective. **Gaussian NB** contributes a probabilistic inductive bias suited to high-dimensional categorical features. A standalone **Logistic Regression** with ℓ_2 regularisation and balanced class weights serves as the interpretable baseline. Diversity among base learners is critical: KNN and NB, though weaker individually (AUC 0.764, 0.755), provide complementary signals with low prediction correlation to tree-based models (mean pairwise $r < 0.85$). The LR meta-learner is trained on out-of-fold (OOF) predictions from inner CV to prevent overfitting, learning to weight each model's strengths across different regions of the feature space (Wolpert 1992).

Nested Cross-Validation. To decouple hyperparameter tuning from generalization estimation, we apply nested 5×5 stratified CV. An inner 5-fold `GridSearchCV` (scoring: AUC-ROC) selects optimal hyperparameters on each outer training partition; the outer fold's held-out set is used *only* for final evaluation. Out-of-fold (OOF) predictions from the four base learners are concatenated into a meta-feature matrix, which trains the meta-classifier. Test predictions are averaged across inner folds to reduce variance before stacking.

Results & Discussion

Table 1: Nested CV AUC-ROC (mean ± std, 5 outer folds).

Model	AUC-ROC	Role
Stacked Ensemble	0.8039 ± 0.0049	**Best**
XGBoost	0.8038 ± 0.0055	Base learner
Random Forest	0.8009 ± 0.0057	Base learner
Logistic Reg.	0.7890 ± 0.0033	Baseline
KNN	0.7636 ± 0.0062	Base learner
Naive Bayes	0.7553 ± 0.0034	Base learner

Model Performance. Table 1 summarises nested CV AUC-ROC. The stacked ensemble achieves $\mathbf{0.8039 \pm 0.0049}$ (95% CI: 0.794–0.814), consistently above the LR baseline (0.789). XGBoost is the strongest individual learner (0.8038), followed by Random Forest (0.8009); KNN and NB are weaker individually but contribute complementary signal to the meta-learner. The stable variance across all five outer folds confirms that the nested framework successfully avoids optimism bias. At the default 0.5 threshold, the baseline LR achieves marginally higher recall (0.640 vs. 0.636), but the ensemble delivers superior precision (0.378 vs. 0.335), reflecting better probability calibration. The ensemble's true advantage emerges under threshold tuning (Figure 2): at profit-maximising thresholds, it achieves both higher precision (0.290 vs. 0.278) and higher recall (0.847 vs. 0.781) simultaneously.

Business Impact. Economic assumptions are grounded in published sources. The **cost per call** of 3.00 reflects nearshore European BPO rates of 1.50–3.00 per completed contact (CallForce Global 2023; LinkedIn Industry Insights 2023). The **revenue per conversion** of 62.50 is derived from a representative 5,000 term deposit (consistent with entry-level Portuguese retail products (ActivoBank 2024)) at Portugal's 2021 average Net Interest Margin of 1.25% (World Bank 2022; Federal Reserve Bank of St. Louis 2022). These parameters give a break-even precision of just 4.8%—comfortably exceeded by all models. Table 2 reports results at profit-maximising thresholds.

Table 2: Business metrics at profit-maximising thresholds.

Strategy	Calls	Net Profit	Calls Saved
Random (all)	39,404	169,163	—
Baseline LR	30,774	177,615	8,630 (21.9%)
Stacked Ensemble	25,125	182,125	**14,279 (36.2%)**

The ensemble saves **14,279 calls** versus random dialling while generating 12,962 more profit (7.6% uplift). Under a resource-constrained scenario (top-10% budget, 3,940 calls), it yields **2,025 conversions** (4.40× lift), versus 1,924 for the LR baseline. The ML system does not replace human telemarketers; rather, it functions as a triage layer ensuring agents engage only with high-probability leads.

Feature Insights. Baseline LR coefficients (Table 5) identify the dominant drivers. The Euribor 3-month rate (-0.734) is the strongest: lower rates reduce the opportunity cost of locking capital, increasing term-deposit appeal (Moro, Cortez, and Rita 2014). Prior contact (*contacted_before*: $+0.381$) is the leading positive predictor. May contact is negative (-0.315), while March, July, and October are positive, pointing to seasonal receptivity. Higher contact frequency (*campaign*: -0.154) and telephone versus cellular contact (-0.137) reduce propensity, suggesting call-fatigue and channel effects. The meta-learner assigns highest weight to XGBoost predictions (0.42), followed by RF (0.31), KNN (0.18), and NB (0.09), reflecting their relative predictive strength.

Societal Considerations. Using demographic features (age, job, marital status) risks systematic disadvantage for groups with historically low conversion rates; regular fairness audits are advisable. Campaign records contain sensitive financial data requiring GDPR-compliant handling. Automated targeting reduces raw call volume, shifting telemarketing roles from high-volume cold calling toward fewer, more specialised sales interactions with pre-qualified leads. Ensemble predictions lack inherent transparency; SHAP-value decomposition should precede regulatory deployment.

Appendix

Methodology Diagram

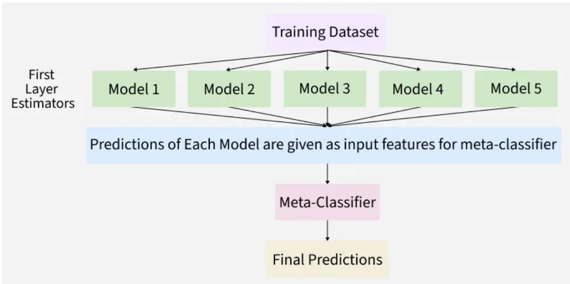

Figure 1: Stacked Generalization architecture. Base learner predictions serve as input features for the Logistic Regression meta-classifier.

Confusion Matrices

Table 3 shows the aggregated confusion matrices at the default 0.5 threshold across all 5 outer folds (39,404 samples total).

Table 3: Confusion matrices at default 0.5 threshold.

		Pred: No	Pred: Yes
Baseline LR	Actual: No	28,977	5,829
	Actual: Yes	1,657	2,941
Stacked Ensemble	Actual: No	29,991	4,815
	Actual: Yes	1,673	2,925

Comprehensive Classification Metrics

Table 4 compares all classification metrics at both the default threshold and at the profit-maximising threshold.

Table 4: Comprehensive classification metrics comparison.

Metric	Baseline LR	Ensemble	Improvement
At default 0.5 threshold			
Accuracy	0.810	0.835	+3.1%
Precision	0.335	0.378	+12.7%
Recall	0.640	0.636	−0.5%
F1-Score	0.440	0.474	+7.8%
AUC-ROC	0.789	0.803	+1.9%
At profit-maximising threshold			
Threshold	0.240	0.272	—
Precision	0.140	0.164	+16.98%
Recall	0.939	0.896	−4.61%
F1-Score	0.244	0.277	+13.51%

Top Feature Importances

Table 5 shows raw feature importance from the baseline Logistic Regression model (not meta-model weights). Positive coefficients increase subscription probability; negative decrease it.

Table 5: Top 10 raw features from baseline Logistic Regression coefficients.

Feature	Coefficient	Interpretation
euribor3m	−0.734	Lower rates → higher appeal
contacted_before	+0.381	Prior contact is positive signal
month_may	−0.315	May contacts less effective
month_mar	+0.287	March highly receptive
month_oct	+0.251	October favourable
month_jul	+0.223	July above average
campaign	−0.154	Call fatigue effect
contact_telephone	−0.137	Cellular more effective
cons.conf.idx	+0.129	Consumer confidence matters
job_student	+0.118	Students more receptive

Threshold and Business Analysis

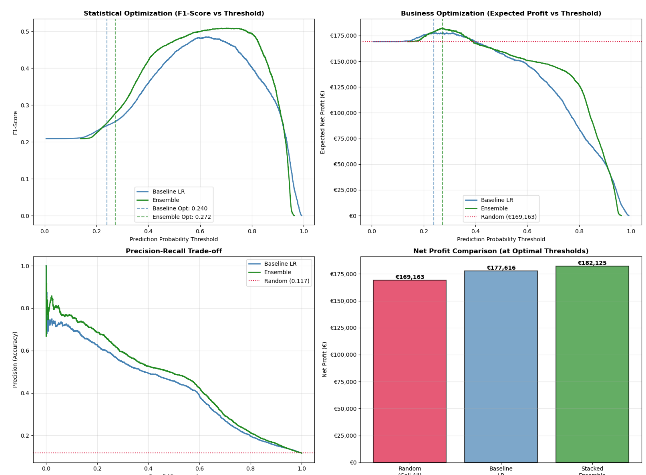

Figure 2: Top left: F1-score vs. classification threshold, showing ensemble peak F1 at higher threshold (0.7) than baseline (0.6). Top right: profit-maximising threshold analysis yields 182,125 while calling only 25,125 prospects. Bottom left: Precision-Recall curve illustrates ensemble maintaining higher precision across all recall levels. Bottom right: direct profit comparison showing 7.6% uplift over baseline.

Ensemble Diversity Analysis

Table 6 shows pairwise prediction correlations between base models, justifying their inclusion in the ensemble despite varying individual performance.

Table 6: Base model prediction correlation matrix (mean across 5 folds). Low correlations between tree-based and distance/probabilistic models support ensemble diversity.

	XGB	**RF**	**KNN**	**NB**
XGB	1.00	0.89	0.76	0.68
RF	0.89	1.00	0.78	0.71
KNN	0.76	0.78	1.00	0.73
NB	0.68	0.71	0.73	1.00

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AI Usage Declaration

Generative AI was utilised to support research, verify the correctness of methodology and analysis, and outline implementation steps. It further assisted in refining code structure, improving written documentation, and enhancing the overall presentation.